

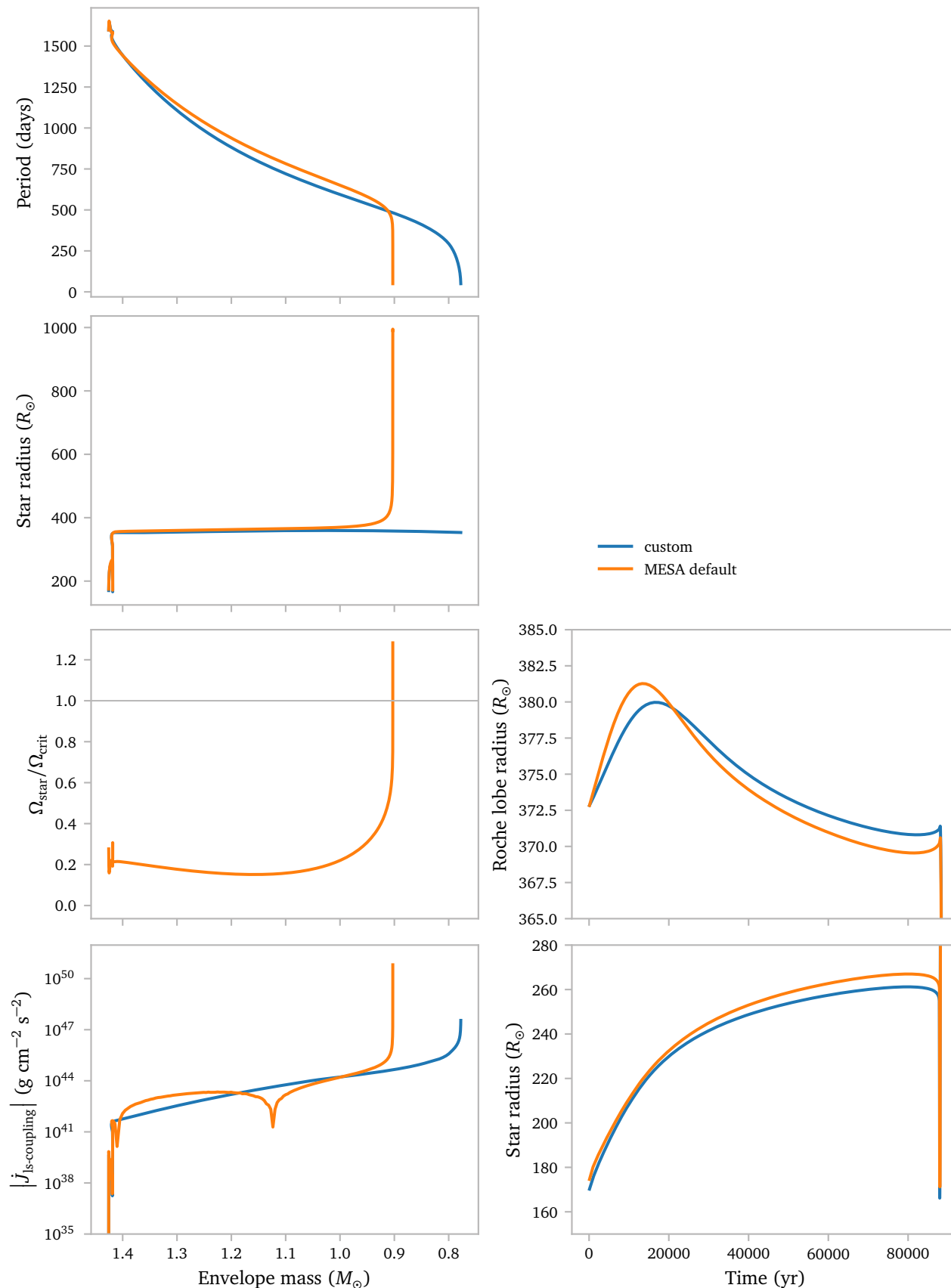
Notes Week 18

a. Mesa Tides

a.1 RLOF test

In order to test the agreement between the standard *MESA* tidal prescription, which is based on *BSE* and the standard [Hut \(1981\)](#) formulism, and my tidal prescription, which is also based on *BSE* and [Hut \(1981\)](#), but with the modifications to the k/T term by [Preece et al. \(2022\)](#).

As a first test, I compare one of the Darwin Instability models from my previous grid with a *MESA* default tides model.



The critical rotation in the bottom left panel is defined as

$$\Omega_{\text{crit}} = \sqrt{\frac{\Gamma GM}{R^3}},$$

where Γ is given by

$$\Gamma \equiv 1 - \frac{L_{\text{rad}}}{L_{\text{edd}}}$$

Unfortunately I do not have the required *MESA* output data to compute this for my custom tides prescription.

There are a couple of interesting things here. First and most importantly, we see the same Darwin instability for the models that use the *MESA* tidal prescription. There are some small differences in the exact strength and effects of the tides though. If anything, the *MESA* tides seem to be a bit stronger than my custom implementation. The differences between the two prescription are in the k/T factor, and the way the convective regions are determined.

The *MESA* rotation causes the star to be slightly more expanded, which makes sense. This has a minor effect on the time at which RLOF can start. The radius also grows significantly once the start gets close to, and eventually exceeds its critical rotation.

This effect is not present in my models, since I do not account for this in any way. It is an interesting effect though, and I am wondering if there might be more models that reach its critical rotation during orbital shrinking. This might affect stability as well.

a.2 Divergence test

Besides looking at the stability of the tides during RLOF, it is also interesting to look at the general differences between the tidal prescription for longer evolutionary periods. Here, I try to compare the differences between the effects of the tides on the orbit during the thermal pulses for my custom prescription and the standard *MESA* implementation.

References

Hut, P. (1981) *Tidal evolution in close binary systems.*. *A&A***99**, 126.

Preece, H. P., Hamers, A. S., Neunteufel, P. G., Schaefer, A. L., & Tout, C. A. (2022) *The Equilibrium Tide: An Updated Prescription for Population Synthesis Codes.* *ApJ***933**, 25.